MGN342-CA2

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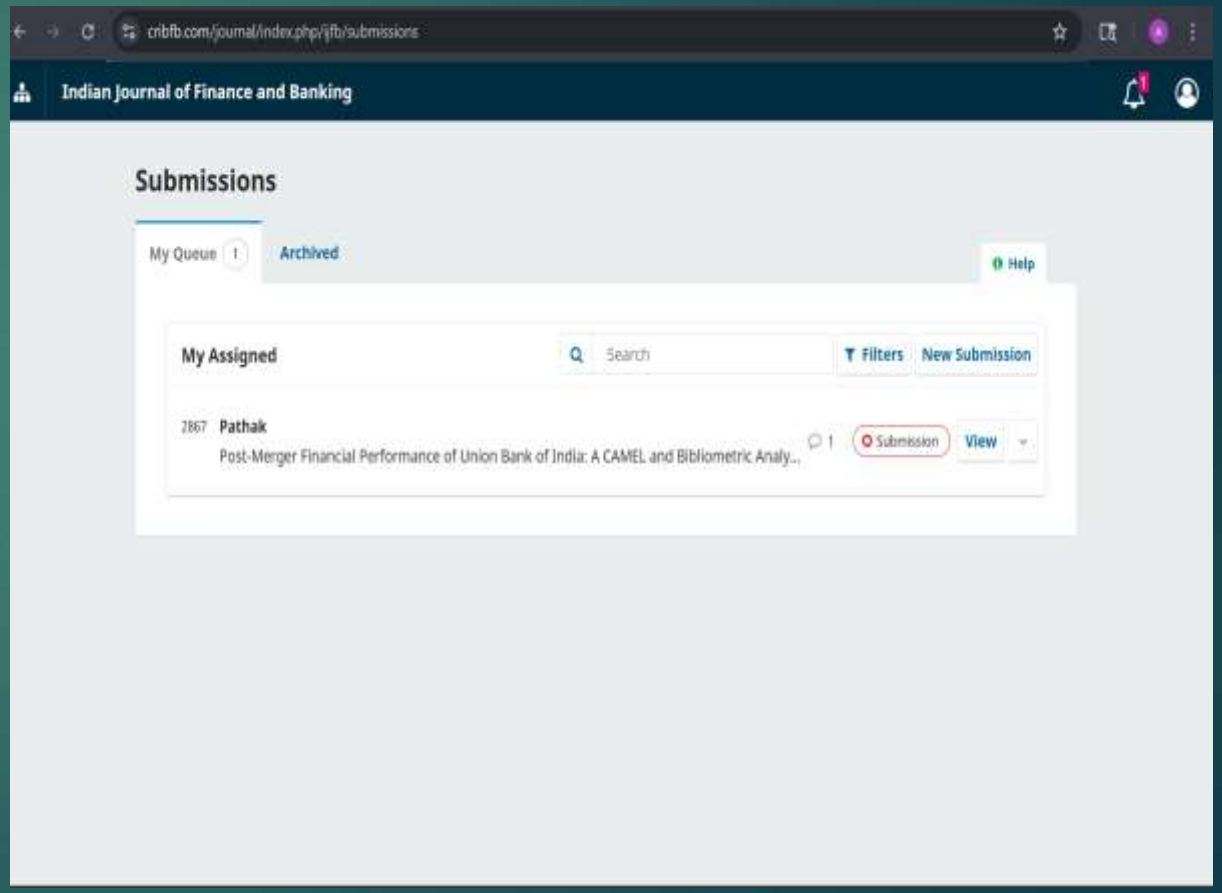
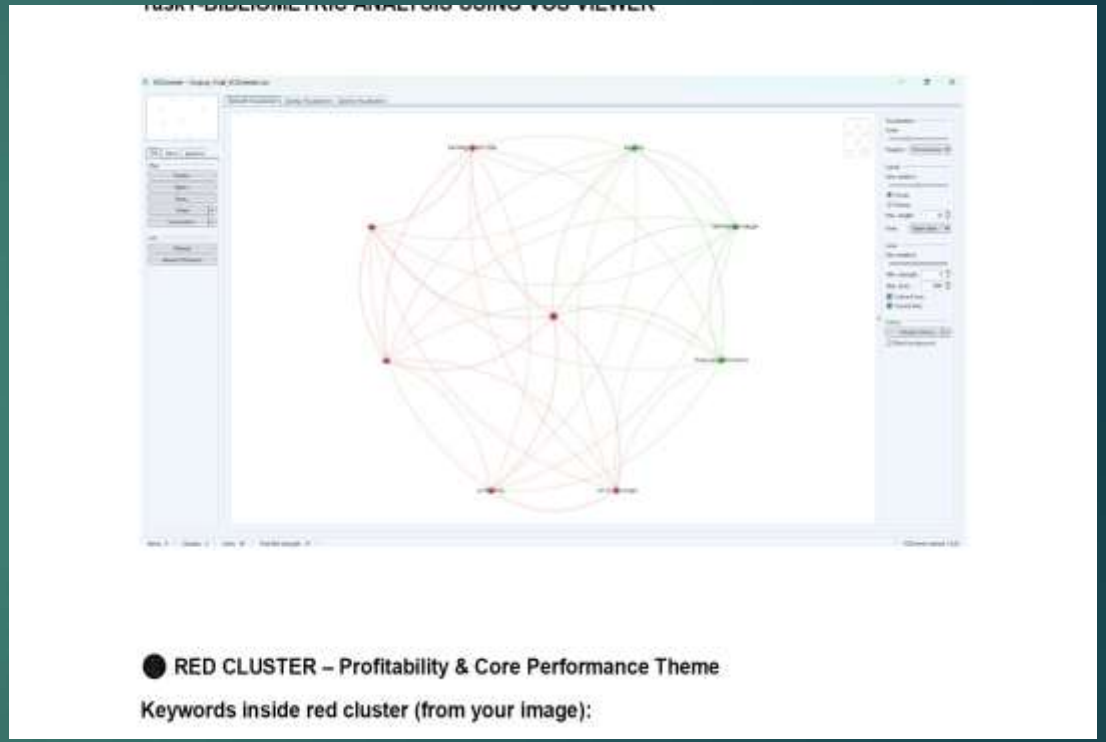
SUBMISSION AND ACCEPTANCE

- SO I HAVE POST MY DOCUMENT ON INDIAN JOURNAL OF FINANCE AND BANKING (IJFB)
- LINK - https://www.cribfb.com/journal/index.php/ijfb/submissions



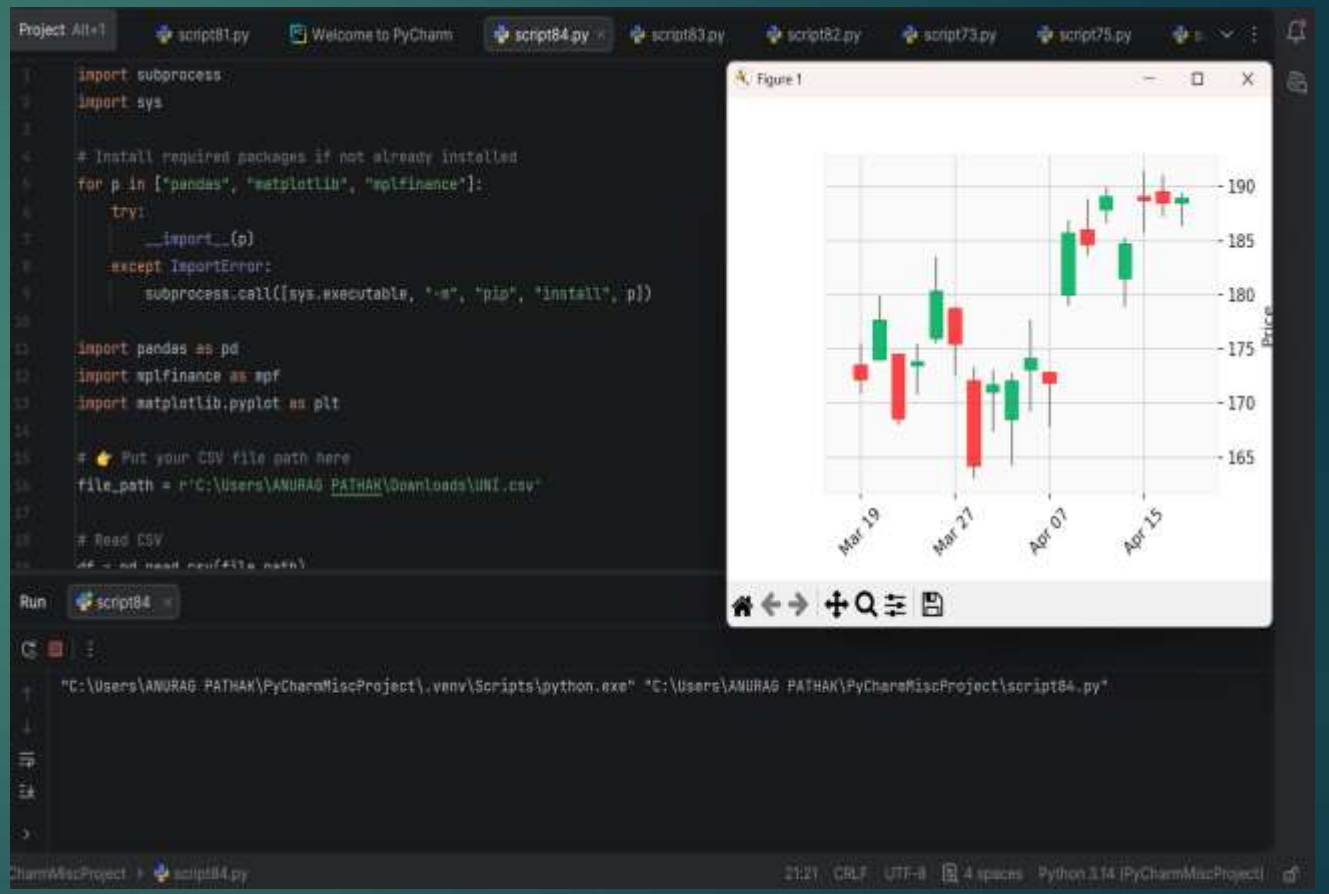
BIBLIOMETRIC IMAGE OF UBI

- **The analysis revealed strong thematic interconnections within banking research, highlighting the integrated nature of performance evaluation, regulatory frameworks, and risk management in the banking sector**



CANDKE STICK CODE

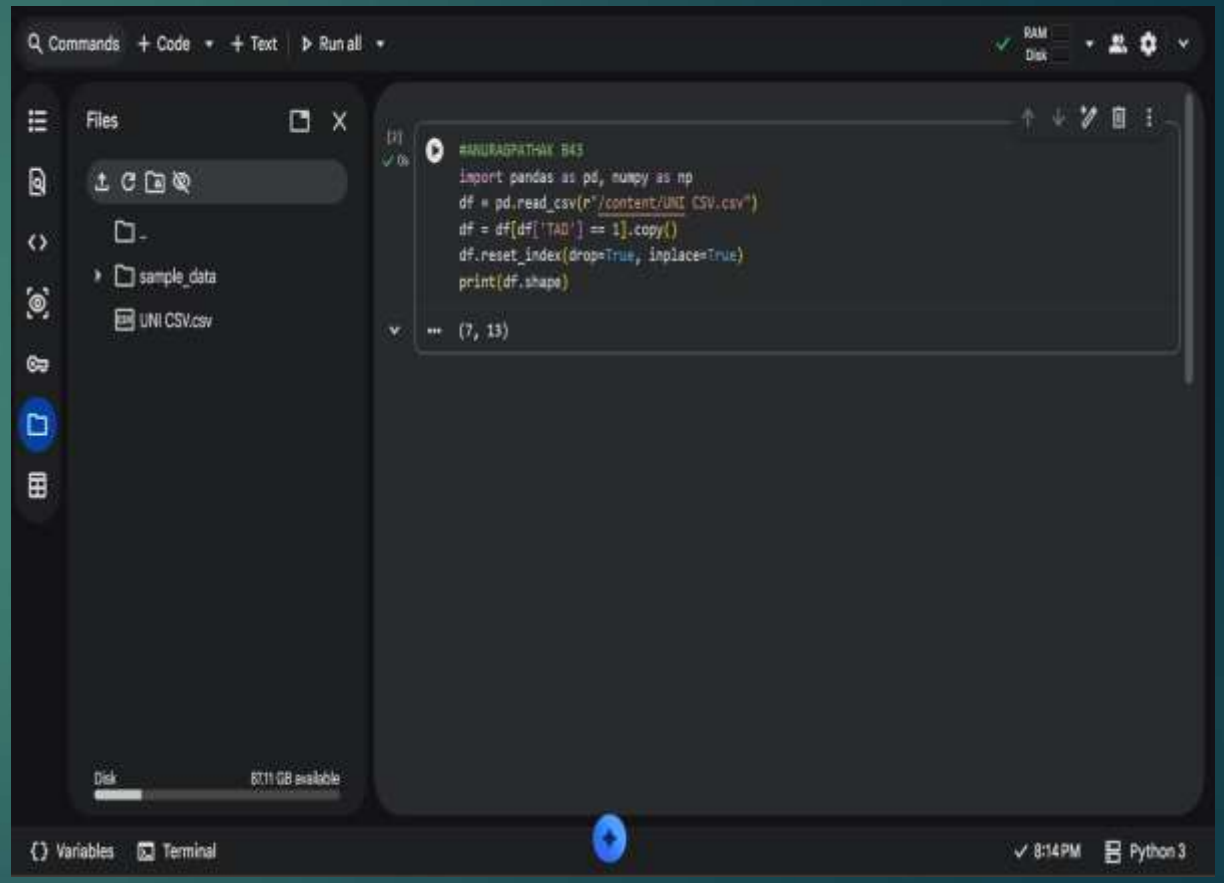
- Candlestick charts are used to analyze price movements in financial markets.Each candle represents four key values: Open, High, Low, and Close.The body of the candle shows the price range between open and close.Green candles indicate a price increase, while red candles indicate a decrease.This chart helps in identifying trends, reversals, and market sentiment.
- **LINKED IN LINK -** https://www.linkedin.com/posts/anuragpathak0724_python-dataanalysis-finance-activity-7451575208521433089-2ryt?utm_source=share&utm_medium=member_android&rcm=ACoAAF2KrXMB20IxSYr5yKzTfoW5rRq5mlufRb0



1.How do we remove rows that failed the attention check?

- `import pandas as pd, numpy as np`
- `df = pd.read_csv(r"C:\D4P\Tatasteel.csv")`

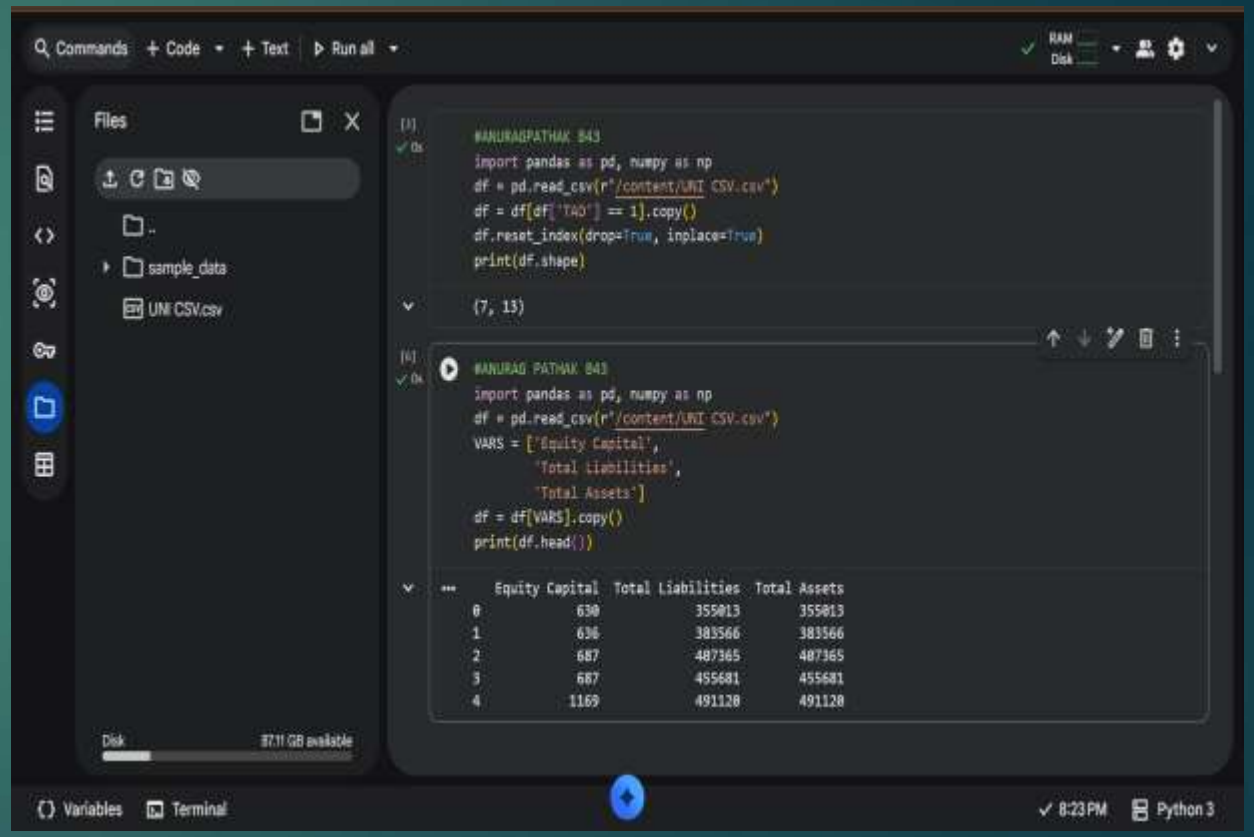
```
df = df[df['TAD'] == 1].copy()

df.reset_index(drop=True, inplace=True)

print(df.shape)
```



2.How do we select only the variables we need?

- import pandas as pd, numpy as np
- df = pd.read_csv(r"C:\D4P\Tatasteel.csv")

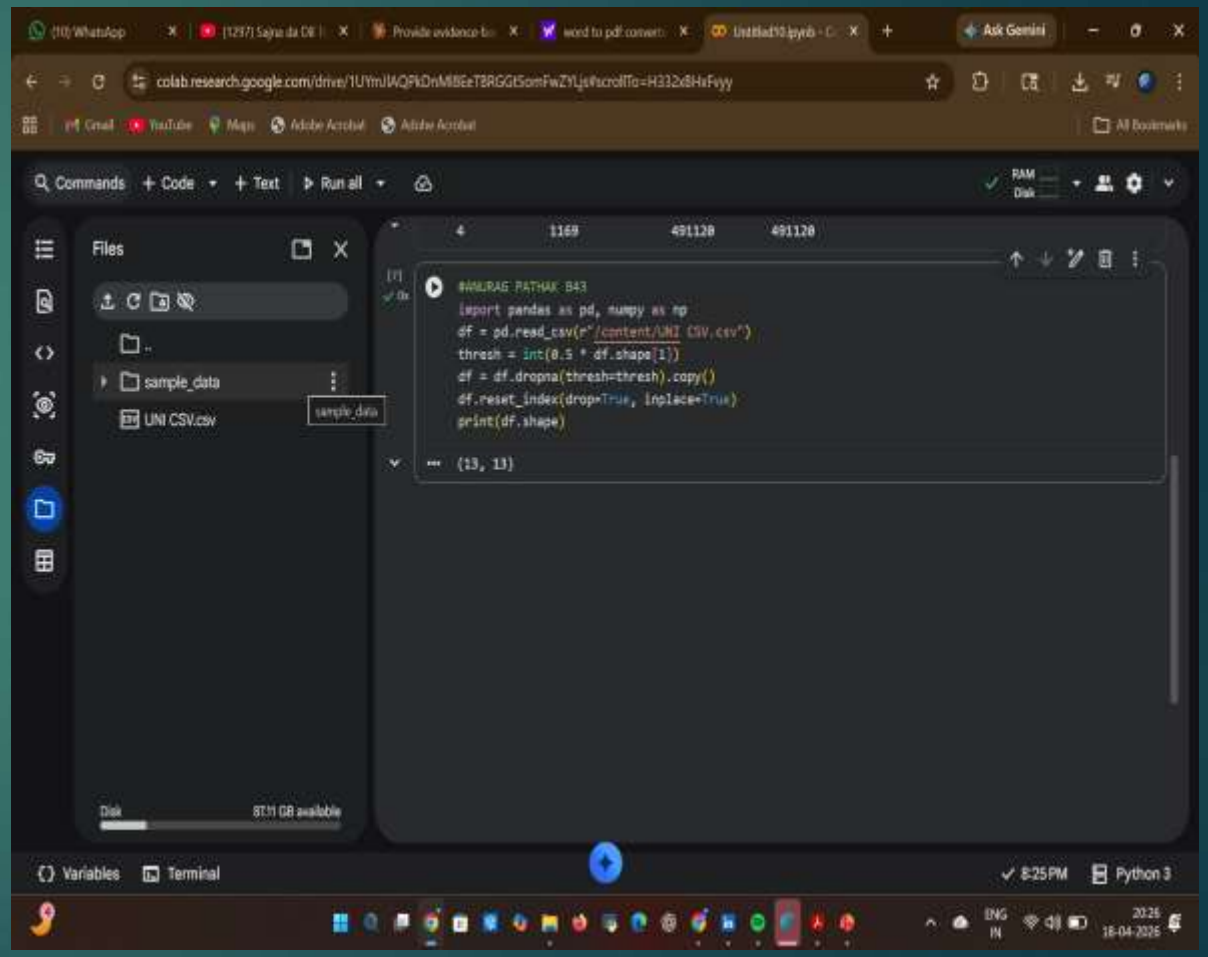
```
VARS = ['EquityCapital',
        'TotalLiabilities',
        'TotalAssets']
df = df[VARS].copy()
print(df.head())
```



3.How do we delete rows where more than 50% of values are missing?

- `import pandas as pd, numpy as np`
- `df = pd.read_csv(r"C:\D4P\Tatasteel.csv")`

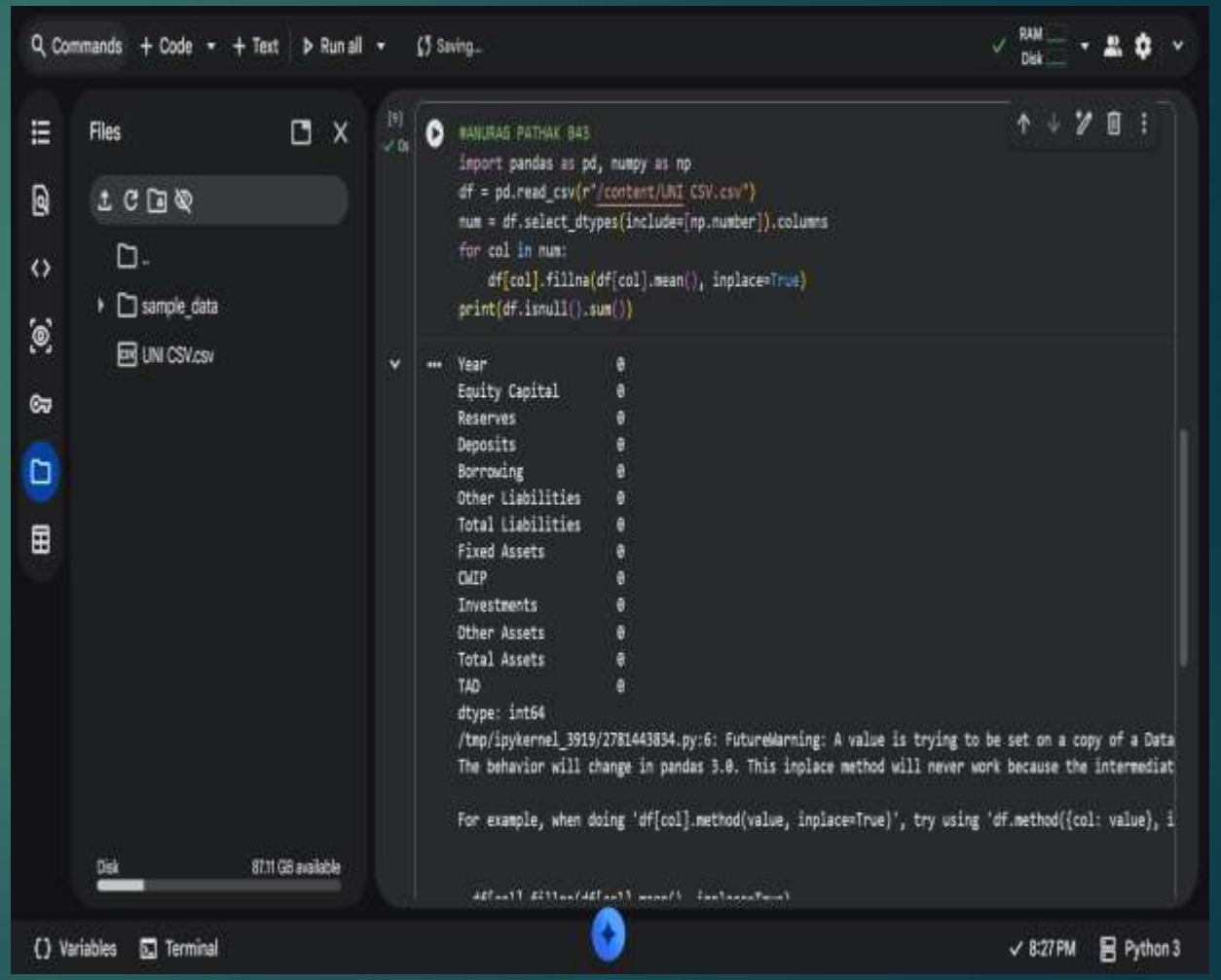
```
thresh = int(0.5 * df.shape[1])

df = df.dropna(thresh=thresh).copy()

df.reset_index(drop=True, inplace=True)

print(df.shape)
```



4.How do we fill missing numeric values with the column mean?

- import pandas as pd, numpy as np
- df = pd.read_csv(r"C:\D4P\Tatasteel.csv")

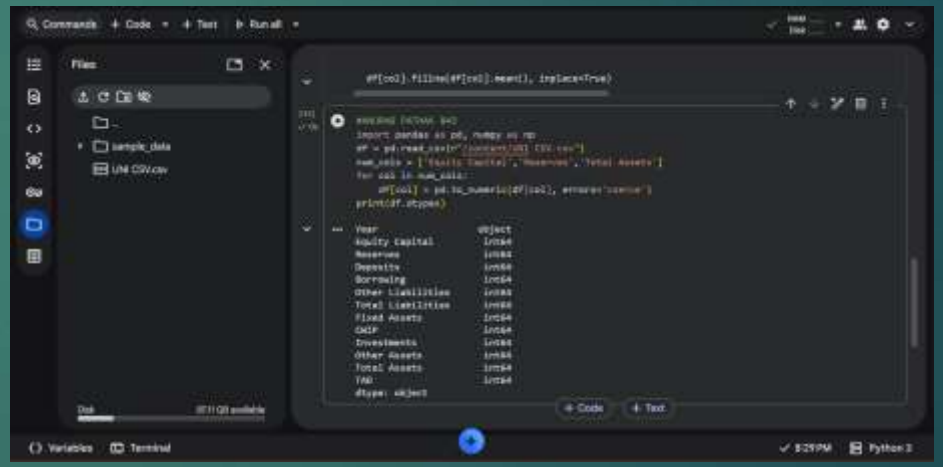
```
num = df.select_dtypes(include=[np.number]).columns

for col in num:


df[col].fillna(df[col].mean(), inplace=True)

print(df.isnull().sum())
```



5.How do we ensure all numeric columns have the correct data type?

- import pandas as pd, numpy as np
- df = pd.read_csv(r"C:\D4P\Tatasteel.csv")

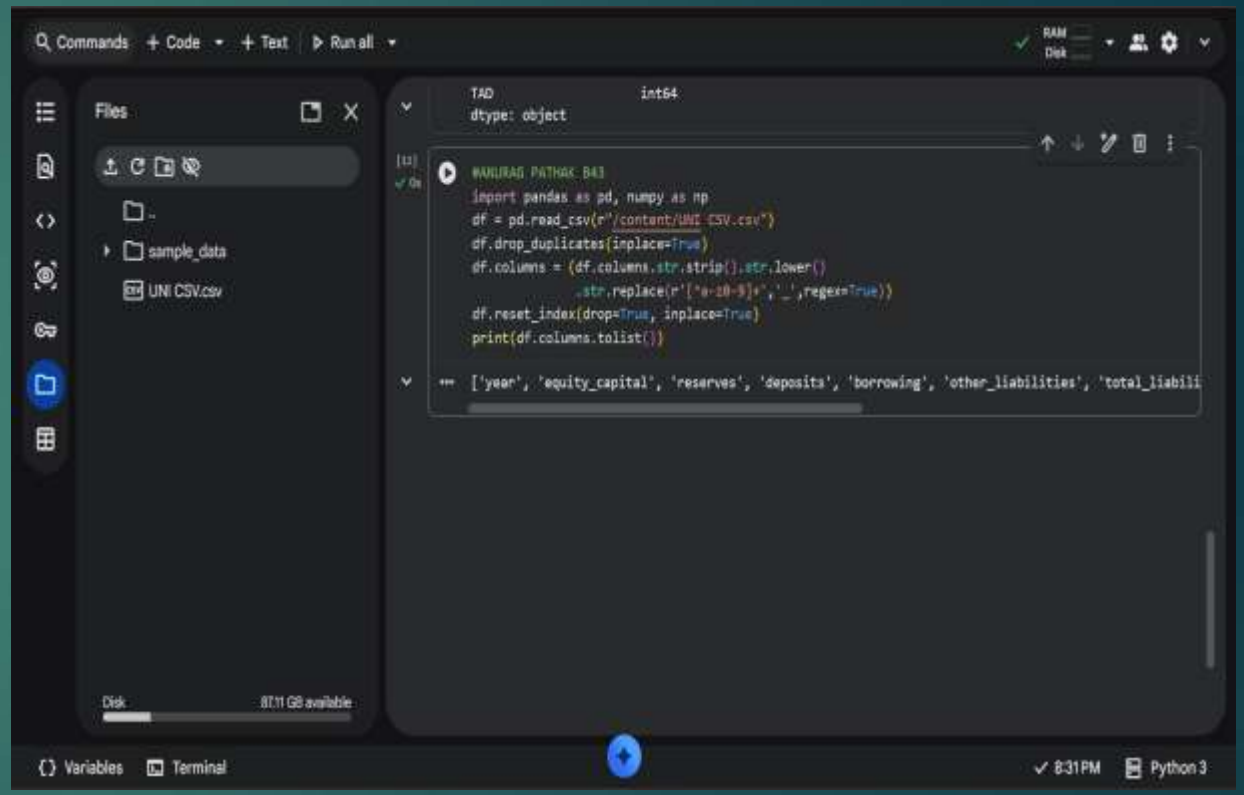
```
num_cols = ['EquityCapital','Reserves','TotalAssets']

for col in num_cols:

    df[col] = pd.to_numeric(df[col], errors='coerce')

print(df.dtypes)
```



6.How do we remove duplicates and standardise column names?

- import pandas as pd, numpy as np
- df = pd.read_csv(r"C:\D4P\Tatasteel.csv")

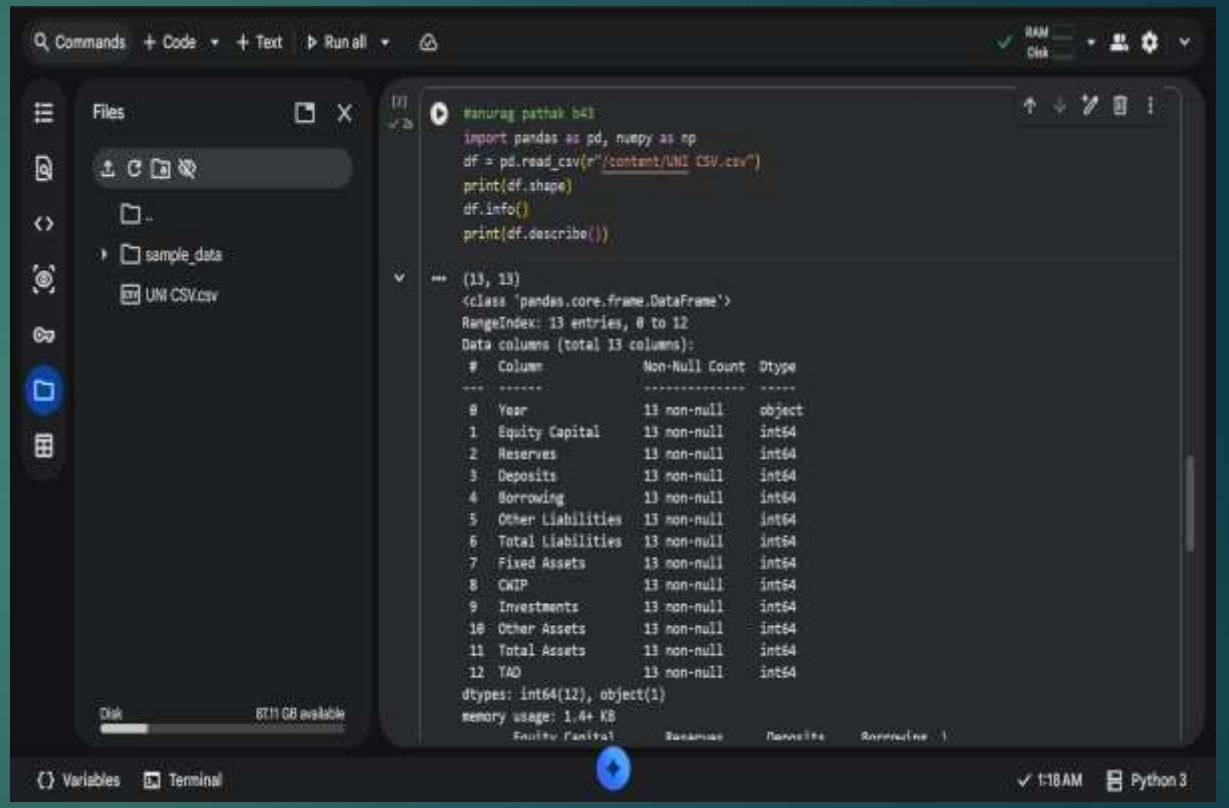
```
df.drop_duplicates(inplace=True)

df.columns =
(df.columns.str.strip().str.lower()

              .str.replace(r'[^a-
z0-9]+','_',regex=True))

df.reset_index(drop=True,
inplace=True)

print(df.columns.tolist())
```



7.How do we get a summary of the dataset's structure?

- `import pandas as pd, numpy as np`
- `df = pd.read_csv(r"C:\D4P\Tatasteel.csv")`

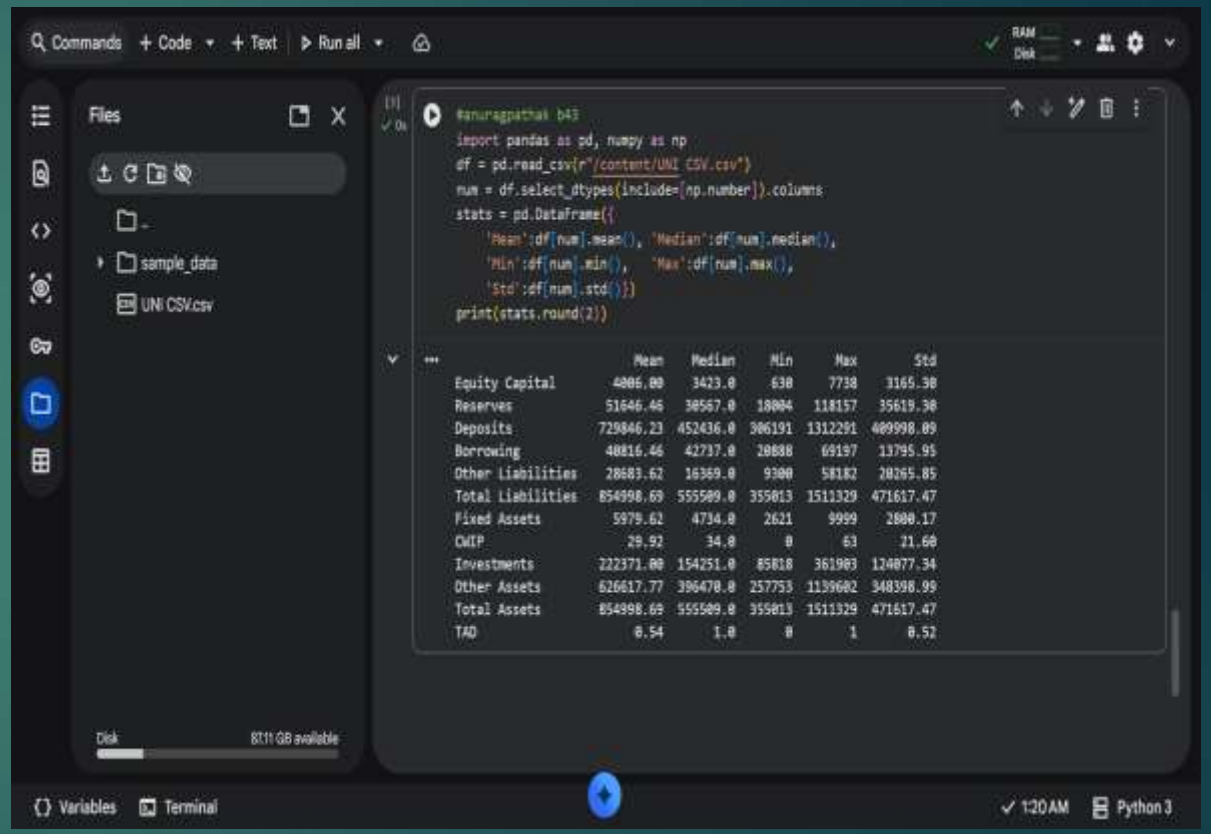
```
print(df.shape)
df.info()
print(df.describe())
```



8.How do we compute mean, median, min, max and std deviation?

- import pandas as pd, numpy as np
- df = pd.read_csv(r"C:\D4P\Tatasteel.csv")

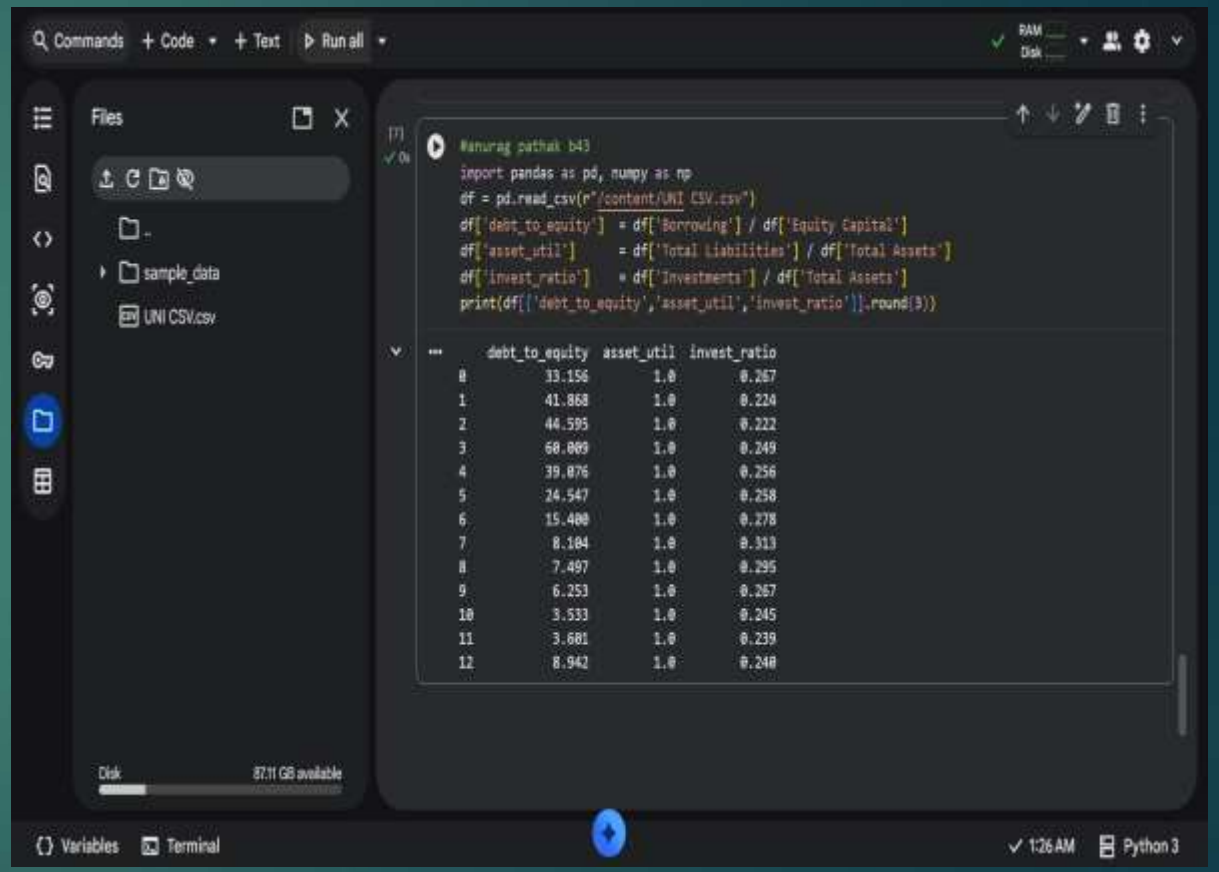
```
num = df.select_dtypes(include=[np.number]).columns

stats = pd.DataFrame({

    'Mean':df[num].mean(),
'Median':df[num].median(),

    'Min':df[num].min(),
'Max':df[num].max(),

    'Std':df[num].std()})

print(stats.round(2))
```



9.How do we create new derived columns from existing ones?

- import pandas as pd, numpy as np
- df = pd.read_csv(r"C:\D4P\Tatasteel.csv")

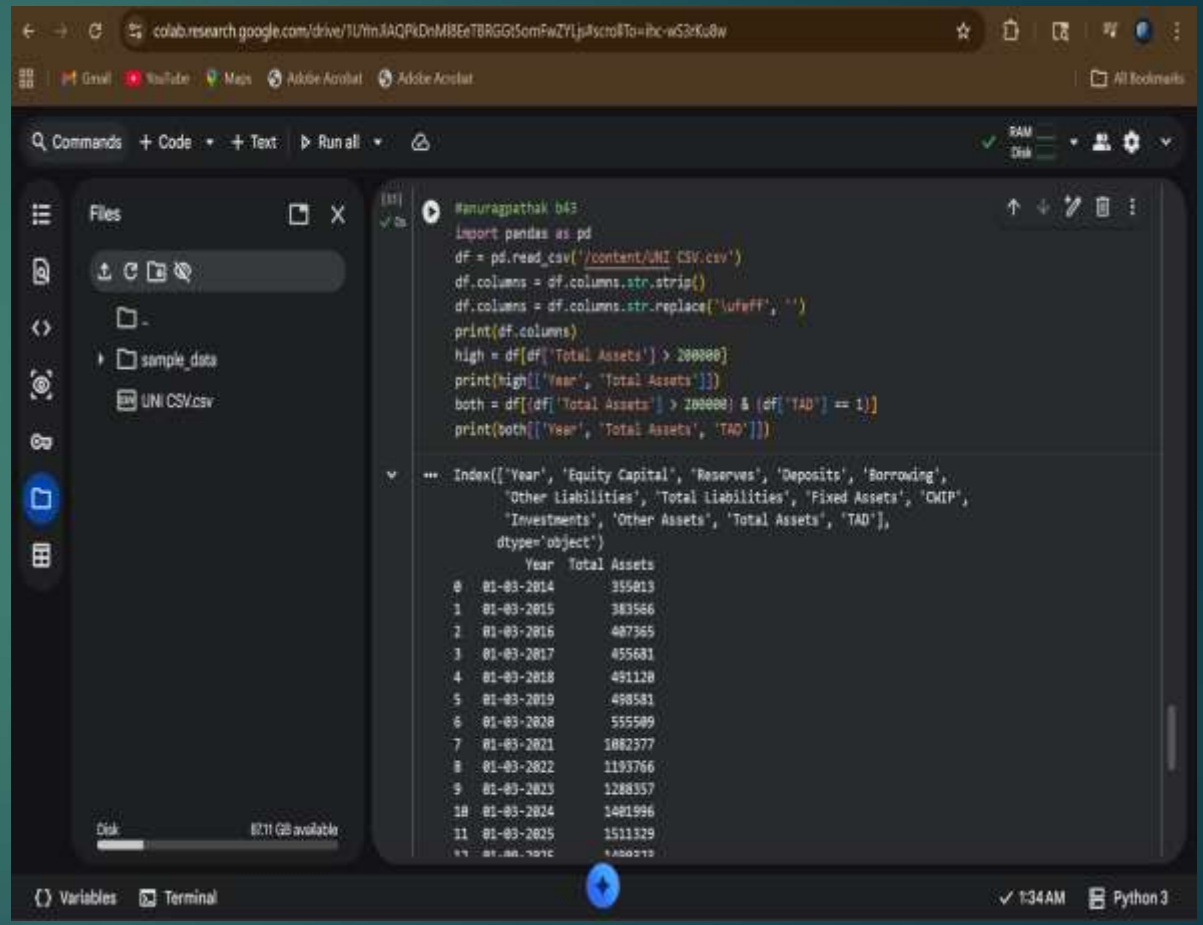
```
df['debt_to_equity']  =
df['Borrowings +'] /
df['EquityCapital']

df['asset_util']      =
df['TotalLiabilities'] /
df['TotalAssets']

df['invest_ratio']    =
df['Investments'] /
df['TotalAssets']

print(df[['debt_to_equity','asset_u
til','invest_ratio']].round(3))
```



10.How do we filter rows based on a condition?

- import pandas as pd, numpy as np
- df = pd.read_csv(r"C:\D4P\Tatasteel.csv")

```
high = df[df['TotalAssets'] > 200000]

print(high[['Year','TotalAssets']])

both = df[(df['TotalAssets'] > 200000) & (df['TAD'] == 1)]

print(both[['Year','TotalAssets','TAD']])
```



11.How do we compute group-wise statistics using groupby?

- `import pandas as pd, numpy as np`
- `df = pd.read_csv(r"C:\D4P\Tatasteel.csv")`

```
grp = df.groupby('TAD')[['EquityCapital', 'TotalLiabilities','TotalAssets']].mean().round(2)

print(grp)
```